BridgeIt

Project Overview (Product Requirements Document) · Version 2.0 · 31 October 2026

A phone game that teaches students how to connect sentences with coordinating conjunctions.

> **How to read this document.** This version replaces Draft v0.3. It follows your Canva screens (“BridgeIt Pre-design”, 39 pages) and your answers in the design review. Items marked **To confirm** still need your decision; they are all collected in Section 11. Items marked **Our proposal** are suggestions you can accept or change. Items marked **Placeholder** are used for building and testing until your final version arrives.

1. What is BridgeIt?

BridgeIt is a game for Android phones. It teaches students the seven coordinating conjunctions, remembered as FANBOYS: *for, and, nor, but, or, yet, so*.

A girl character is going on a road trip and needs the student’s help to reach her destination. Along the way, the student learns the conjunctions through a song, repairs broken bridges by joining two clauses with the right conjunction, strengthens the bridges with sentences of their own, and finally writes a travelogue about a place in Ilocos Norte.

The game is used in class as part of a study led by teacher-researchers. The teacher gives a code that opens each next level, and the students send their answers to the teacher for the study.

2. What students will learn

After playing all three levels, the student should be able to:

1. Remember the seven FANBOYS conjunctions and what each one does.
2. Use them to join two independent clauses correctly, with the right punctuation.
3. Write sentences of their own with each conjunction and with words of similar meaning.
4. Use coordinating conjunctions in a short piece of writing (a travelogue entry).

Not included in this version

- Teaching other kinds of conjunctions. Words such as *because* and *however* appear only as similar words in the Quick Facts card (Section 8).
- Student accounts, logins, online saving or a leaderboard.
- A version for iPhone or for web browsers.

3. Who will use it

Who	What they need
Students	A simple, encouraging game with clear feedback and a chance to try again. Grade level: **To confirm**
Teacher-researchers	Control over when students move to the next level, using codes. A report of each student’s answers and results for the study (Section 7).

4. Where it runs

- **Device:** Android phones, held upright.
- **Delivery:** an Android app file (APK) installed on the students' phones.
- **Language:** English only.
- **Internet:** not needed. The app works fully offline after it is installed. Students do not sign in.
- **Sound:** a button turns the sound on and off. **Our proposal**

5. How the game flows

1. **Start screen.** The BridgeIt logo and a play button.
2. **Welcome.** The character greets the student ("Hi, Bridge Builder!") and explains the journey. The student types their name and section once; these appear on their answer report.
3. **Level menu.** Shows the three levels. Levels 2 and 3 stay locked until the student enters the teacher's code in the menu. Completed levels show a tick.
4. **Levels 1 to 3**, described in Section 6.
5. **Finish.** After Level 3, the character thanks the student and the journey ends.

A small map icon at the top of each level screen shows the journey so far: the levels done and where the car is.

A finished level can be played again for practice. **Our proposal:** the answer report keeps the student's first completed attempt, so replays do not change the research data.

6. The three levels

Level 1: Conjunction Theme Song

Listen and sing along

- The character asks the student to listen to the Conjunction Theme Song three times.
- A song player shows the progress bar and a play counter (1/3, 2/3, 3/3).
- A sing-along card shows the lyrics line by line in time with the song, with a picture for each part.
- MatchIt opens after the third full play.

MatchIt

- An instruction card asks "Are you ready?" with two buttons: "Yes!" and "No. Let me listen to the song again."
- The student matches each of the seven conjunctions to its function (Section 8). They can drag a function into its slot, or tap a function and then tap the slot.
- On **Submit**, right pairs stay in place and wrong ones go back to the bottom for the student to try again.
- The student can go back to the song at any time without playing it three times again.
- When all pairs are right, the level is done and the teacher gives the Level 2 code.

Level 2: Building Bridges

The level has **7 bridges, one for each FANBOYS conjunction**, on one continuous road. Each bridge has three steps.

Step 1: Build the bridge

- Two independent clauses sit on either side of a broken bridge (for example, “She went to bed early” and “she was tired.”).
- The seven pieces (*, for*, *, and*, *, nor*, *, but*, *, or*, *, yet*, *, so*) are shown below the bridge. The student drags a piece into the gap and taps **Cross the bridge**.
- **Wrong answer:** a message explains what that conjunction does (for example, “‘so’ is used to show result or consequence.”), and the piece fades out of the tray.
- **After 3 wrong tries** the bridge collapses. That bridge restarts with all pieces back; the rest of the level is not repeated.
- **Right answer:** a message explains why the conjunction fits (for example, “‘she was tired’ explains why ‘She went to bed early’, so we use ‘for’.”).

Step 2: Strengthen the bridge with your own sentence

- The student types their own sentence using the same conjunction.
- The app checks the sentence (Section 7). If it passes, the sentence is added under the bridge as a new foundation. If not, the app shows what to fix and the student tries again.

Step 3: Quick Facts and a second sentence

- The student taps the conjunction to open the Quick Facts card, which lists words with a similar meaning and how to punctuate them (Section 8).
- The student chooses one of those words and writes a second sentence with it. If it passes the checks, it becomes a second foundation.
- *nor* and *or* have no similar words, so those two bridges have Step 2 only.

In total the student writes **12 sentences** in this level: two for each of *for, and, but, yet, so*, and one each for *nor* and *or*.

When the bridge is sturdy, the car drives across to the next bridge. Progress is saved after each bridge, so a student can stop and continue later. After the seventh bridge the road reaches the toll gate, and the teacher gives the Level 3 code.

Level 3: Travelogue

- **Toll gate.** The character arrives at the toll gate and passes through. This is a story scene; the Level 3 code is entered in the level menu.
- **Map.** A map of Ilocos Norte shows seven places. The student chooses **one**:
  1. Laoag La Paz Sand Dunes
  2. Burgos Lighthouse (Cape Bojeador)
  3. Bangui Windmills
  4. San Agustin Church of Paoay
  5. Saud Beach, Pagudpud
  6. La Virgen Milagrosa de Badoc Church
  7. Pangil Coral Rock Formations, Currimao
- **Montage.** The chosen place plays a short montage of **5 pictures**, each with a caption, shown one after another with a slow zoom and fade, over simple background music. The captions are listed in Appendix A.
- **Instructions.** The character says goodbye and asks the student to write about the trip: where they went, what they did and saw, how they felt and what they learned.
- **Travelogue entry.** The student writes **5 to 7 sentences** with **at least 4 coordinating conjunctions** that join two clauses, without other help from the app. A counter shows progress, and **Submit** opens

when both targets are met.

- **Ending.** The character congratulates the student and the journey ends.

7. What the app will do

Checking the students' writing

Every sentence a student writes in Level 2 is checked before it can become part of the bridge. Each sentence must:

- use the right word: the coordinating conjunction, or one of the similar words from Quick Facts;
- have the right punctuation for that word: a comma before a FANBOYS word, a semicolon before and a comma after words like *however*, and no comma needed before words like *because*;
- have a full clause on both sides, each with a subject and a verb. "I like apples, and she likes oranges." passes; "I like apples, and oranges." does not;
- start with a capital letter and end with . ! or ?;
- not be copied from the example on screen or from the student's earlier sentences.

If a check fails, the app shows one short message saying what to fix. When the app is not sure whether a part is a full clause, it shows a hint but lets the student continue, and the teacher has the final say.

In the travelogue, a conjunction only counts towards the minimum of 4 when it has a comma before it and joins two full clauses. Counted conjunctions are highlighted as the student types.

Spelling warning. Misspelled words are underlined so the student can fix them. They do not block the student.

What the app does not check. Grammar, meaning and word choice. For example, "She go to school, for she want to learn." passes the checks. The teacher reviews these in the answer report.

Answer report for the teacher and researchers

After each level, and at the end of the game, the student taps **Send my answers**. The app creates a **PDF report** and opens the phone's share menu, so the student can send it to the teacher by Messenger, email, Google Drive, Bluetooth or Nearby Share. The app itself stays offline.

The report includes everything the student has done so far:

Part	Contents
Student	Name, section, date and the app version
Level 1	MatchIt result: number of Submit attempts, and which pairs were wrong on each attempt. Number of song plays. Time spent
Level 2	For each of the 7 bridges: wrong tries, how many times the bridge collapsed, time spent, and each written sentence with the number of attempts it took and any spelling warnings
Level 3	The place chosen, the full travelogue entry, the number of sentences and counted conjunctions, spelling warnings and time spent
Totals	Time spent on each level and in total

Time spent counts only while the level is open on screen; it pauses when the student leaves the app.

Checklist

- Start screen, welcome with name and section, and a level menu where Levels 2 and 3 open only with the teacher's code.
- Remember the student's progress, answers and results, even after the app is closed.
- Level 1: song with play counter and sing-along card, then MatchIt with drag or tap.
- Level 2: 7 bridges with drag-and-drop, fading wrong pieces, collapse after 3 wrong tries, and sentence writing with checks and spelling warnings.
- Level 3: toll gate, map, montage with captions and music, and the travelogue with live counters.
- Map icon showing the journey so far.
- Send my answers: PDF report shared through the phone's share menu.
- Sound on and off.
- Let the team change sentences, hints, similar words, captions, limits and codes without changing the app's code.

8. Learning content

Conjunction	What it does (from MatchIt)	Similar words (Quick Facts)	Punctuation of similar words
for	Expresses a reason or explanation	because, since, as	No comma before
and	Adds or connects related ideas	moreover, in addition, furthermore	Semicolon before, comma after
nor	Connects negative alternatives	None	–
but	Shows contrast between ideas	however, although	*however:* semicolon before, comma after. *although:* no comma before
or	Presents choices or alternatives	None	–
yet	Shows an unexpected result	nevertheless, even though	*nevertheless:* semicolon before, comma after. *even though:* no comma before
so	Expresses a result or consequence	therefore, as a result	Semicolon before, comma after

Level 2 sentences. The 7 sentence pairs, with the explanation for each right answer and a hint for each wrong piece: **To confirm** (you will send these). Until then the app uses **Placeholder** sentences, starting with "She went to bed early, for she was tired." from your screens.

9. Look and feel

- **Screens:** built as in your Canva design: the character, the car, the driver's-seat view with speech boxes, the bridge scenes, the map and the place scenes.
- **Logo:** your BridgeIt logo on the start, welcome and finish screens, and as the app icon.
- **Colors:** taken from your logo. The values below are measured from the logo image and are approximate until you confirm the exact codes.

Color	Code (approx.)	Used for
Navy	#183870	Text, main buttons, headers
Teal	#309090	Cards, buttons, bridge details
Light teal	#80C0A8	Speech boxes, highlights
Orange	#D88858	Conjunction pieces, accents

Right and wrong answers are also shown with icons, words and movement, not only color.

Text corrections we will make in the screens (Our proposal): “acticity” → “activity”; “can not” → “cannot”; “has given” → “has been given”; “LightHouse” → “Lighthouse”; “will to” → “will too”; “Travelouge” → “Travelogue”; “Building Bridges” used as the Level 2 title everywhere; “Bridge Builder” used everywhere.

10. Sound

- **Conjunction Theme Song:** your recording, with the time each lyric line starts for the sing-along card. **To confirm** (you will send these).
- **Montage music:** simple, calm background music. **Placeholder:** a free track with a license that allows use in apps. If you send your own music, we will use it instead.
- **Sound effects** (for example, a bridge cracking or a car driving): **To confirm**

11. Questions and items we need from you

#	Question or item	Status
1	What grade level are the students?	To confirm
2	Will students play with the teacher guiding the class, on their own, or both?	To confirm
3	The unlock codes for Levels 2 and 3, or should the teacher set them?	Expected Saturday
4	The 7 Level 2 sentence pairs, with explanations and hints (Section 8)	Expected Saturday
5	Captions for the 35 montage pictures. Please replace or approve our placeholders in Appendix A	To confirm
6	The 35 montage pictures and the other art as separate files (PNG or SVG), with characters and backgrounds separate	Expected Saturday
7	The song as an MP3 file, and the lyrics with the time each line starts	Expected Saturday
8	Background music for the montage, or use our free placeholder track	To confirm
9	Should the game have sound effects besides the song and music?	To confirm
10	The logo as an SVG or a transparent PNG at least 1024 px wide, a square icon-only version (bridge and book, no text), and the exact color codes	To confirm

12. Changes from Draft v0.3

- Level 1: fill-in-the-blanks replaced by a sing-along card and MatchIt.
- Level 2: 7 bridges, each strengthened with sentences the student writes, checked by the app.
- Level 3: the same-meaning word activity is replaced by the Travelogue.
- New: answer report (PDF) for the teacher and researchers, spelling warnings, map icon, name and section.
- Look and feel: your logo, character and Canva screens replace our proposed lotus logo and wireframes.

13. Attachments and references

- Your Canva design “BridgeIt Pre-design” (39 pages).
- The design review document with your answers.
- Draft v0.3 of this document (BridgeIt-PRD.pdf) and the earlier wireframes (BridgeIt-Wireframes.pdf), now replaced.

Appendix A. Montage captions (Placeholder)

These captions are placeholders written by the development team so the montage can be built and tested. Please check the facts, then replace or approve each one.

1. Laoag La Paz Sand Dunes

1. The La Paz Sand Dunes stretch along the coast of Laoag City.
2. Visitors ride 4x4 jeeps up and down the dunes.
3. Sliding down the slopes on a sandboard is a favorite activity.
4. The wind shapes the sand into waves and ridges.
5. At sunset, the dunes turn gold and orange.

2. Burgos Lighthouse (Cape Bojeador)

1. Cape Bojeador Lighthouse stands on a hill in Burgos.
2. It was first lit in 1892, during the Spanish period.
3. A brick path leads up to the lighthouse tower.
4. From the hill, visitors can see the wide blue sea.
5. The lighthouse still guides ships at night.

3. Bangui Windmills

1. The Bangui Windmills line the shore of Bangui Bay.
2. Twenty giant wind turbines stand in a row.
3. The turbines turn the strong sea wind into electricity.
4. Waves crash on the rocky beach below the windmills.
5. The windmills look beautiful against the evening sky.

4. San Agustin Church of Paoay

1. San Agustin Church in Paoay is a UNESCO World Heritage Site.
2. It was finished in 1710 and built of coral stone and bricks.
3. Its huge side buttresses help it stand strong during earthquakes.
4. A separate bell tower stands beside the church.

5. Visitors walk around the grounds to admire its old walls.

5. Saud Beach, Pagudpud

1. Saud Beach in Pagudpud has fine white sand.
2. Coconut trees line the shore.
3. The clear blue water is calm and good for swimming.
4. Families have picnics in the shade of the trees.
5. In the afternoon, the beach is quiet and peaceful.

6. La Virgen Milagrosa de Badoc Church

1. The church in Badoc is home to La Virgen Milagrosa.
2. A legend says fishermen found the image of the Virgin floating at sea.
3. The church has thick walls and an old stone facade.
4. Many people visit to pray and light candles.
5. The church stands at the heart of the town of Badoc.

7. Pangil Coral Rock Formations, Currimao

1. Coral rock formations line the coast of Currimao.
2. Waves and wind have shaped the rocks over many years.
3. At low tide, visitors can walk among the rocks and tide pools.
4. Small sea creatures live in the pools between the rocks.
5. At sunset, the rocks glow in warm colors.